

Final Submission Report

Paddy Research Agent — Agentic AI Research Support Tool

Research domain: Lightweight CNNs vs Hybrid Vision Transformers for Paddy Disease Detection on Edge Devices in South Asian Agriculture.

1. Executive Summary

This project implements Option B, a Research Support Tool around the registered research topic. It plans research questions, retrieves domain evidence, synthesizes a grounded report and applies a separate review pass.

2. Requirements Mapping

Requirement	Implementation
≥3 agentic patterns	Planner–Executor, Tool-using RAG, Reflection/Critic
≥2 agents	ResearchPlanner, EvidenceAnalyst, QualityReviewer
Agent communication	Pydantic AgentMessage with sender, recipient, type, payload and trace ID
≥2 models	OpenRouter model for planning/synthesis and Groq model for review
RAG	24 domain-specific source cards + TF-IDF retrieval
Retrieval evaluation	5 fixed queries with Precision@5 benchmark
Streamlit	app.py provides workflow, trace, evidence and reviewer views
Secrets	Environment variables / Streamlit secrets
GitHub	15-commit feature-branch workflow documented; student must perform remote PRs
README	Setup, architecture, models, RAG, evaluation, limitations and demo instructions

3. Agent Architecture

ResearchPlanner creates a structured research plan. EvidenceAnalyst uses the retrieval tool and drafts an evidence-grounded response. QualityReviewer checks citation coverage, unsupported claims and limitations.

4. RAG Design

The corpus contains 24 curated source cards covering paddy datasets, plant disease datasets, lightweight CNNs, Vision Transformers, hybrid architectures, edge deployment, optimization and explainability. Retrieval uses TF-IDF and cosine similarity.

5. Evaluation

Five fixed queries are included. The evaluation script reports Precision@5 and the retrieved source IDs; manual relevance assessment should be shown in the demonstration.

6. Model Orchestration

Planning and evidence synthesis use OpenRouter by default, while review uses Groq. Model IDs are configurable through environment variables. Actual provider cost and latency must be recorded from the student's final run.

7. Security

Only environment variables or Streamlit secrets are used for API keys. No real secrets belong in Git.

8. Limitations

The corpus is not a systematic review; TF-IDF is a lightweight retrieval baseline; LLM review does not guarantee correctness; and real edge latency/model-size results must be experimentally measured.

9. Demonstration

Use docs/DEMO_SCRIPT.md. Configure secrets, run the Streamlit app, capture screenshots, record actual model IDs and run the retrieval benchmark before final submission.